

#This code will be used to demonstrate proposition 6.16. We demonstrate that the component nj and nk are positive along the curve γ_2

Define nj

$$nj := (x^{-3} - w^{-3}) \cdot w + (x^{-3} - z^{-3}) \cdot z;$$

Define nk

$$nk := (w^{-3} - x^{-3}) \cdot (y^{-3} - z^{-3}) + (z^{-3} - x^{-3}) \cdot (y^{-3} - w^{-3})$$

$$nj := \left(\frac{1}{x^3} - \frac{1}{w^3} \right) w + \left(\frac{1}{x^3} - \frac{1}{z^3} \right) z$$

$$nk := \left(\frac{1}{w^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{z^3} \right) + \left(\frac{1}{z^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{w^3} \right) \quad (1)$$

#Apply the rational parametrization constructed in this chapter 4.

$$parametrization := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$x = a - \frac{1}{a} + b - \frac{1}{b},$$

$$y = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(numer(expand(subs(parametrization, nj))));

simplify(numer(expand(subs(parametrization, nk))));

$$\begin{aligned} & - (b^2 + 1) (b + a) a (a^2 + 1) (a^9 b^5 + (2 b^6 - 3 b^4) a^8 + (b^7 - 4 b^5 + 3 b^3) a^7 + (2 b^8 + b^6 \\ & - 6 b^4 - 5 b^2) a^6 + (b^9 - 4 b^7 - 20 b^5 + 2 b^3 + 3 b) a^5 + (-3 b^8 - 6 b^6 + 12 b^4 - 1) a^4 \\ & + (3 b^7 + 2 b^5 - 3 b^3 - 2 b) a^3 + (-5 b^6 - 3 b^2) a^2 + (3 b^5 - 2 b^3) a - b^4) b \end{aligned}$$

$$\begin{aligned} & 12 \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1) a^8}{2} \right. \\ & + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1) a^6}{6} \\ & - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1) a^4}{2} \\ & \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) a^3 b^3 \quad (2) \end{aligned}$$

Define $\tilde{nj}$

$$\tilde{nj} := (a^9 b^5 + (2 b^6 - 3 b^4) a^8 + (b^7 - 4 b^5 + 3 b^3) a^7 + (2 b^8 + b^6 - 6 b^4 - 5 b^2) a^6 + (b^9$$

$$-4b^7 - 20b^5 + 2b^3 + 3b)a^5 + (-3b^8 - 6b^6 + 12b^4 - 1)a^4 + (3b^7 + 2b^5 - 3b^3 - 2b)a^3 + (-5b^6 - 3b^2)a^2 + (3b^5 - 2b^3)a - b^4):$$

Define ñk

$$\begin{aligned} \text{ñk} := & \left((b^8 - b^6 + b^4)a^{10} - \frac{b^2(b^8 - 2b^6 + 8b^4 - 2b^2 + 1)a^8}{2} \right. \\ & + \frac{b(b^4 - 4b^2 + 1)(b^2 + 1)^3a^7}{2} + \frac{(b^4 + 1)(b^8 + 8b^4 + 1)a^6}{6} \\ & - \frac{b(b-1)^2(b+1)^2(b^2 + 1)^3a^5}{2} - \frac{b^2(b^8 + 2b^6 - 2b^4 + 2b^2 + 1)a^4}{2} \\ & \left. - \frac{b^3(b^2 + 1)^3a^3}{3} + a^2b^6 - \frac{b^6}{3} \right): \end{aligned}$$

#Define r2

$$\begin{aligned} r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4b^{18} + 8b^{16} - 4b^{14})a^2 + \right. \\ & -4b^{18} + 8b^{16} - 4b^{14})a^{30} + (6b^{20} - 32b^{18} + 44b^{16} - 32b^{14} + 6b^{12})a^{28} + b^{16} - 9 \left(b^{16} \right. \\ & + \frac{4168}{9}b^{14} - 528b^{12} - 328b^{10} + \frac{3698}{9}b^8 - 72b^6 - 16b^4 + 8b^2 + 1 \left. \right) b^8 a^{24} - 4b^2 (b^{28} \\ & + 12b^{26} + 37b^{24} + 378b^{22} + 4187b^{20} + 19656b^{18} + 41514b^{16} + 57358b^{14} + 65514b^{12} \\ & + 34120b^{10} + 4315b^8 + 1018b^6 - 155b^4 + 12b^2 + 1)a^{14} - 4b^2 (b^{28} + 76b^{26} - 539b^{24} \\ & - 2118b^{22} + 5211b^{20} + 36360b^{18} + 92394b^{16} + 78414b^{14} + 38058b^{12} + 17992b^{10} - 2085b^8 \\ & - 518b^6 + 37b^4 + 12b^2 + 1)a^{18} + 8 \left(b^{20} + b^{18} - 699b^{16} + 2369b^{14} - \frac{1037}{2}b^{12} - 1827b^{10} \right. \\ & + \frac{2163}{2}b^8 - 159b^6 - 27b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024b^9 (b^2 + 1)^4 a^7 + 8b^6 \left(b^{20} + b^{18} \right. \\ & - 27b^{16} + 33b^{14} - \frac{2189}{2}b^{12} - 3875b^{10} - \frac{9357}{2}b^8 - 3231b^6 - 2747b^4 - 1055b^2 - 127 \left. \right) \\ & a^{10} + 2b^4 (b^{24} + 36b^{22} + 180b^{20} - 1372b^{18} - 9036b^{16} - 25400b^{14} - 54798b^{12} - 64184b^{10} \\ & - 27212b^8 - 6620b^6 - 2636b^4 + 36b^2 + 1)a^{12} + (-1024b^{22} + 72b^{20} + 692b^{18} + 8b^{16} \\ & - 76b^{14} + 72b^{12})a^{26} + (-254b^{28} + 840b^{26} + 6504b^{24} - 696b^{22} - 25752b^{20} - 169584b^{18} \\ & - 181788b^{16} - 85104b^{14} - 50328b^{12} - 4792b^{10} + 360b^8 + 72b^6 + 2b^4)a^{20} + (b^{32} + 8b^{30} \\ & + 20b^{28} + 1672b^{26} + 1950b^{24} - 32376b^{22} - 157780b^{20} - 317080b^{18} - 279672b^{16} \end{aligned}$$

$$\begin{aligned}
& - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\
& - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
& + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
& - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
& - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\
& \left. + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):
\end{aligned}$$

#Plot the curves $r_1=0$, $\tilde{n}_j=0$ and $\tilde{n}_k=0$.

$r := a - b :$

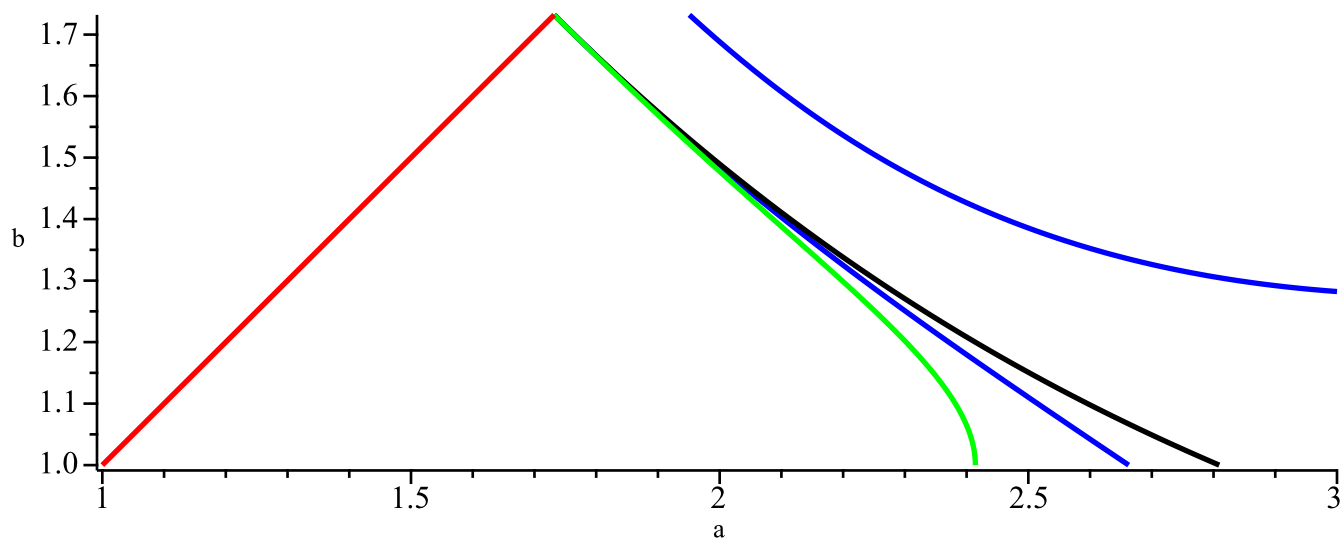
with(plots) :

```

implicitplot([r2=0, ñj=0, ñk=0, r=0], a=1..3, b=1..(sqrt(3)),
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, black, green, red],
  labels=["a", "b"],
  title="Graphic for r2=0, ñj=0 and ñk=0 ");

```

Graphic for $r_2=0, \tilde{n}_j=0$ and $\tilde{n}_k=0$



#Compute the resultant between r_2 and $\tilde{n}_j$ with respect to a

$Ra := \text{resultant}(r_2, \tilde{n}_j, a) :$

#Compute the factors of the resultant Ra and count how many there are

$fator := [\text{op}(\text{factor}(Ra))] :$
 $\text{nops}(fator);$

8 (3)

#We list below the factors of the resultant Ra . When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[1, \sqrt{3})$ where 12 is defined

$fator[1];$

65536 (4)

$fator[2];$

b^{144} (5)

$fator[3];$

$\text{sturm}(\text{sturmseq}(fator[3], b), b, 1, (\sqrt{3})); \# \text{number of roots in } (1, (\sqrt{3}))]$

$$81 b^{16} + 432 b^{14} + 9540 b^{12} - 13536 b^{10} + 44686 b^8 + 2800 b^6 + 1108 b^4 + 192 b^2 + 9$$

0 (6)

$fator[4];$

$\text{sturm}(\text{sturmseq}(fator[4], b), b, 1, (\sqrt{3})); \# \text{number of roots in } (1, (\sqrt{3}))]$

$$\begin{aligned} &3969 b^{32} + 104328 b^{30} + 1023408 b^{28} + 900936 b^{26} - 21818700 b^{24} + 245450664 b^{22} \\ &+ 6399720480 b^{20} + 31859784168 b^{18} + 58790686846 b^{16} + 41509040792 b^{14} \\ &+ 14661620592 b^{12} + 2161519832 b^{10} + 271691620 b^8 + 35381304 b^6 + 4619328 b^4 \\ &+ 419832 b^2 + 21609 \end{aligned}$$

0 (7)

$fator[5];$

$(b^2 - 3)^2$ (8)

$fator[6];$

$\text{sturm}(\text{sturmseq}(fator[6], b), b, 1, (\sqrt{3})); \# \text{number of roots in } (1, (\sqrt{3}))]$

$$(7 b^4 - 6 b^2 + 3)^2$$

0 (9)

$fator[7];$

$\text{sturm}(\text{sturmseq}(fator[7], b), b, 1, (\sqrt{3})); \# \text{number of roots in } (1, (\sqrt{3}))]$

$$\begin{aligned} &(45241 b^{48} - 983136 b^{46} + 8765868 b^{44} - 53430800 b^{42} + 292102794 b^{40} - 1327146624 b^{38} \\ &+ 4932365356 b^{36} - 14813913168 b^{34} + 36572797215 b^{32} - 74566977600 b^{30} \\ &+ 126486144024 b^{28} - 177535429920 b^{26} + 204746625068 b^{24} - 190367984256 b^{22} \end{aligned}$$

$$\begin{aligned}
&+ 140202896376 \, b^{20} - 79729549344 \, b^{18} + 34666181511 \, b^{16} - 11137697376 \, b^{14} \\
&+ 2222925500 \, b^{12} - 25011408 \, b^{10} - 137706774 \, b^8 + 38273024 \, b^6 - 2821284 \, b^4 \\
&- 1082256 \, b^2 + 374017)^2 \\
&\hspace{15em} 0 \hspace{10em} \textbf{(10)}
\end{aligned}$$

$$\begin{aligned}
&fator[8]; \\
&\hspace{15em} (b^2 + 1)^{66} \hspace{10em} \textbf{(11)}
\end{aligned}$$

#The curves $r_2=0$ and $\tilde{n}j=0$ do not intersect.

#Check the sign of $\tilde{n}_j$ at the point $(\gamma_l(1), 1)$

#Locate a range for $\gamma_l(1)$

sturm(sturmseq(eval(r_2 , $b = 1$), a), $a, 2, 2.7$); #number of roots in $[2, 3)$

1

(12)

#Verify that in this interval $\tilde{n}_j(a, 1)$ has a constant sign.

sturm(sturmseq(eval($\tilde{n}_j$, $b = 1$), a), $a, 2, 2.7$); #number of roots in $[2, 3)$

0

(13)

#Calculate the sign of $\tilde{n}_j(\gamma_l(1), 1)$ using the sign of $\tilde{n}_j(2, 1)$.

eval(eval($\tilde{n}_j$, $b = 1$), $a = 2$)

-831

(14)

#The sign of n_j is opposite to that of $\tilde{n}_j$, it follows that $n_j > 0$ along γ_l

restart

Define ñk

$$\begin{aligned} \tilde{n}k := & \left((b^8 - b^6 + b^4) a^{10} - \frac{b^2 (b^8 - 2b^6 + 8b^4 - 2b^2 + 1) a^8}{2} \right. \\ & + \frac{b (b^4 - 4b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8b^4 + 1) a^6}{6} \\ & - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2b^6 - 2b^4 + 2b^2 + 1) a^4}{2} \\ & \left. - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right) : \end{aligned}$$

#Define r2

$$\begin{aligned} r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4b^{18} + 8b^{16} - 4b^{14}) a^2 + \right. \\ & -4b^{18} + 8b^{16} - 4b^{14}) a^{30} + (6b^{20} - 32b^{18} + 44b^{16} - 32b^{14} + 6b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\ & + \frac{4168}{9} b^{14} - 528b^{12} - 328b^{10} + \frac{3698}{9} b^8 - 72b^6 - 16b^4 + 8b^2 + 1 \left. \right) b^8 a^{24} - 4b^2 (b^{28} \\ & + 12b^{26} + 37b^{24} + 378b^{22} + 4187b^{20} + 19656b^{18} + 41514b^{16} + 57358b^{14} + 65514b^{12} \\ & + 34120b^{10} + 4315b^8 + 1018b^6 - 155b^4 + 12b^2 + 1) a^{14} - 4b^2 (b^{28} + 76b^{26} - 539b^{24} \\ & - 2118b^{22} + 5211b^{20} + 36360b^{18} + 92394b^{16} + 78414b^{14} + 38058b^{12} + 17992b^{10} - 2085b^8 \\ & - 518b^6 + 37b^4 + 12b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699b^{16} + 2369b^{14} - \frac{1037}{2} b^{12} - 1827b^{10} \right. \\ & + \frac{2163}{2} b^8 - 159b^6 - 27b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024b^9 (b^2 + 1)^4 a^7 + 8b^6 \left(b^{20} + b^{18} \right. \\ & - 27b^{16} + 33b^{14} - \frac{2189}{2} b^{12} - 3875b^{10} - \frac{9357}{2} b^8 - 3231b^6 - 2747b^4 - 1055b^2 - 127 \left. \right) \\ & a^{10} + 2b^4 (b^{24} + 36b^{22} + 180b^{20} - 1372b^{18} - 9036b^{16} - 25400b^{14} - 54798b^{12} - 64184b^{10} \\ & - 27212b^8 - 6620b^6 - 2636b^4 + 36b^2 + 1) a^{12} + (-1024b^{22} + 72b^{20} + 692b^{18} + 8b^{16} \\ & - 76b^{14} + 72b^{12}) a^{26} + (-254b^{28} + 840b^{26} + 6504b^{24} - 696b^{22} - 25752b^{20} - 169584b^{18} \\ & - 181788b^{16} - 85104b^{14} - 50328b^{12} - 4792b^{10} + 360b^8 + 72b^6 + 2b^4) a^{20} + (b^{32} + 8b^{30} \\ & + 20b^{28} + 1672b^{26} + 1950b^{24} - 32376b^{22} - 157780b^{20} - 317080b^{18} - 279672b^{16} \\ & - 222104b^{14} - 125524b^{12} - 7288b^{10} - 3426b^8 - 120b^6 + 20b^4 + 8b^2 + 1) a^{16} + (-9b^{24} \end{aligned}$$

$$\begin{aligned}
& - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
& + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
& - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
& - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\
& \left. + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):
\end{aligned}$$

#Compute the resultant between r_2 and $\tilde{n}k$ with respect to a

$Ra := \text{resultant}(r_2, \tilde{n}k, a) :$

#Compute the factors of the resultant Ra and count how many there are

$fator := [op(\text{factor}(Ra))] :$
 $nops(fator);$

7 (15)

#We list below the factors of the resultant Ra . When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[1, \text{sqrt}(3))$ where 12 is defined

$fator[1];$

$$\frac{1}{31088519960728128454656} \tag{16}$$

$fator[2];$

$$b^{160} \tag{17}$$

$fator[3];$

$\text{sturm}(\text{sturmseq}(fator[3], b), b, 1, (\text{sqrt}(3))) ; \# \text{number of roots in } (1, (\text{sqrt}(3)))]$

$$\begin{aligned}
& 201545550647774641 b^{156} + 26091219170754511998 b^{154} + 361585232998046982939 b^{152} \\
& - 130963328751372183668 b^{150} - 130388126939757400959735 b^{148} \\
& - 279967648023527012610558 b^{146} + 3620019864154595920678667 b^{144} \\
& + 160940106010701027705944424 b^{142} - 1972979746709284943883794829 b^{140} \\
& + 13927305220149044325187283410 b^{138} - 79539059828434235052427832607 b^{136} \\
& + 310481324910885734882601124524 b^{134} + 182598334338368271586785687811 b^{132}
\end{aligned}$$

$$\begin{aligned}
& - 12005386098267790709457670903890 \, b^{130} + 92300599687783329815049589360713 \, b^{128} \\
& - 494978415415112481078763544152128 \, b^{126} \\
& + 2023145351439782210888740989477924 \, b^{124} \\
& - 4173537101566204561731174651183240 \, b^{122} \\
& - 13032733636063165568183302585300468 \, b^{120} \\
& + 171791709178326296999845457915208048 \, b^{118} \\
& - 1039795400253525481034862310322193468 \, b^{116} \\
& + 4829981885821823924585546414792734088 \, b^{114} \\
& - 17468220296985996262470337578279187572 \, b^{112} \\
& + 46121682190537227842779749333610917216 \, b^{110} \\
& - 81107980734204901315267408424455543716 \, b^{108} \\
& + 76388760308187455911174939910611108968 \, b^{106} \\
& - 12072987991564623810576421275785035212 \, b^{104} \\
& + 116614354222940452510620157173133006896 \, b^{102} \\
& - 1058590461372895421771282653974865740708 \, b^{100} \\
& + 3540004390856713390340444922603703221912 \, b^{98} \\
& - 7013659464620873378334148195447331743852 \, b^{96} \\
& + 8833375032775702807948649709388563447360 \, b^{94} \\
& - 6563805327143655203940509170540028090466 \, b^{92} \\
& + 3955205261570191183123912384193755367108 \, b^{90} \\
& - 14856122074723924124298860722562802725814 \, b^{88} \\
& + 55979376886560926403656740401524260235304 \, b^{86} \\
& - 128483772924084915191410765495527023342930 \, b^{84} \\
& + 205307506061458741821964537288964359204668 \, b^{82} \\
& - 243591856351288543568598444641772571843158 \, b^{80} \\
& + 220004269000606531937914985420979881281136 \, b^{78} \\
& - 153561585682384177912187927176499510824182 \, b^{76} \\
& + 90729161554414712574897887505486629206332 \, b^{74} \\
& - 61656051419883747450238738450383134563058 \, b^{72} \\
& + 58724410174258598649006211920296209780776 \, b^{70} \\
& - 54063291285060043601451079668741746632662 \, b^{68} \\
& + 35824619471723942177739204744538576767172 \, b^{66} \\
& - 14564401603684848684541945022041763295618 \, b^{64} \\
& + 4732008090683601101198916412388181802560 \, b^{62}
\end{aligned}$$

$$\begin{aligned}
& - 5360558687346575145805290233225679935404 b^{60} \\
& + 7256092189817071284720504677788303708824 b^{58} \\
& - 4966509761132987187123479749849535262948 b^{56} \\
& + 1536436454887031164160118447625589712944 b^{54} \\
& + 15866249334274782087646245588033249780 b^{52} \\
& + 301516206764056666239464248319853242472 b^{50} \\
& - 572304726582372003943456557444324658660 b^{48} \\
& + 343615675526010254474949876055978241376 b^{46} \\
& - 51063565384779734300946344512014119220 b^{44} \\
& - 22149057370604876964509659678447189624 b^{42} \\
& - 12357330403805598669075164837520603900 b^{40} \\
& + 20983044594507402382571162605982323056 b^{38} \\
& - 10762446129766802461316300362871686068 b^{36} \\
& + 3003570392540101865087764748741303160 b^{34} \\
& - 486560265839604073196474197116043164 b^{32} \\
& + 14506644890019352043843463903049664 b^{30} \\
& + 8323582120659525168536263620251961 b^{28} \\
& - 619860369954219114028234260172626 b^{26} - 111426996676113263095944985685901 b^{24} \\
& + 9900984422511556818979295917740 b^{22} - 49863654734008661388943836783 b^{20} \\
& + 162340076637417985627564578514 b^{18} - 13829661974246732966859355485 b^{16} \\
& - 1956213362565512717830833816 b^{14} + 216048149459477528028020827 b^{12} \\
& + 17407565191158737689970946 b^{10} - 1764554824835978005223463 b^8 \\
& - 34068413508632845696116 b^6 - 1002344030045036839125 b^4 \\
& + 347072952894808621950 b^2 - 4170153564935809983
\end{aligned}$$

$$0 \quad (18)$$

fator[4];

$$(b^2 - 3)^2 \quad (19)$$

fator[5];

sturm(*sturmseq*(*fator*[5], *b*), *b*, 1, (sqrt(3))); #number of roots in (1, (sqrt(3))]

$$(7b^4 - 6b^2 + 3)^2$$

$$0 \quad (20)$$

fator[6];

sturm(*sturmseq*(*fator*[6], *b*), *b*, 1, (sqrt(3))); #number of roots in (1, (sqrt(3))]

$$(45241b^{48} - 983136b^{46} + 8765868b^{44} - 53430800b^{42} + 292102794b^{40} - 1327146624b^{38}$$

$$\begin{aligned}
&+ 4932365356 \, b^{36} - 14813913168 \, b^{34} + 36572797215 \, b^{32} - 74566977600 \, b^{30} \\
&+ 126486144024 \, b^{28} - 177535429920 \, b^{26} + 204746625068 \, b^{24} - 190367984256 \, b^{22} \\
&+ 140202896376 \, b^{20} - 79729549344 \, b^{18} + 34666181511 \, b^{16} - 11137697376 \, b^{14} \\
&+ 2222925500 \, b^{12} - 25011408 \, b^{10} - 137706774 \, b^8 + 38273024 \, b^6 - 2821284 \, b^4 \\
&- 1082256 \, b^2 + 374017)^2
\end{aligned}$$

$$0 \tag{21}$$

*f*ator[7];

$$\begin{aligned}
&(b^2 + 1)^{60} \\
&0
\end{aligned}
\tag{22}$$

#The curves $r_2=0$ and $\tilde{n}k=0$ do not intersect.

#Check the sign of $\tilde{n}k$ at the point $(\gamma_l(1), 1)$

#Locate a range for $\gamma_l(1)$

sturm(sturmseq(eval(r_2 , $b = 1$), a), a , 2.5, 2.7); #number of roots in $[2, 3)$

1

(23)

#Verify that in this interval $\tilde{n}k(a, 1)$ has a constant sign.

sturm(sturmseq(eval($\tilde{n}k$, $b = 1$), a), a , 2.5, 2.7); #number of roots in $[2, 3)$

0

(24)

#Calculate the sign of $\tilde{n}j(\gamma_l(1), 1)$ using the sign of $\tilde{n}k(2.5, 1)$.

eval $\left(\text{eval}(\tilde{n}k, b = 1), a = \frac{5}{2} \right)$

2384551
3072

(25)

γ_2 lies to the right of $\tilde{n}k$